# SAMYAK JAIN

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## Professional Summary

Software Engineer experienced in building backend services, event-driven systems, frontend applications, shared developer tooling, and cloud infrastructure for a multi-tenant K–12 EdTech platform. Strong foundation in API design, distributed systems, data modeling, object-oriented programming, and data structures and algorithms.

## Experience

**Compro Technologies — Delhi** *Jan 2025 – Present*

*Software Engineer*

**Project: K–12 LMS (School Framework) — multi-tenant platform supporting 1M+ active users across schools**

- Built the core Roster Service to model **6 OneRoster entities** -- organizations, users, courses, classes, enrollments, and academic sessions -- per the 1EdTech OneRoster data standard, with dedicated Lambda handlers per entity to process incoming roster data.
- Integrated Rostersteam (Ednition) as the roster ingestion source, consuming OneRoster-formatted CSV data through event-driven Lambda handlers to **automate onboarding** across multiple schools.
- Built a Calendar Service end to end -- event modeling, ingestion, and calendar-date processing following the EdFi standard schema -- using a **CQRS-based read model** for user-course aggregations and progress tracking, improving read throughput by roughly **35%** while supporting multi-tenancy.
- Architected and delivered an **event-driven transcript generation platform** using AWS SQS, Lambda, DynamoDB, API Gateway, and S3, maintaining **99.9% availability**.
- Reduced frontend load times by cutting **FCP and LCP by ~0.5s** via image optimization, lazy loading, and code splitting; and refactored **15+ legacy modules** into a reusable component library, improving maintainability across shared applications.
- Fixed a **cross-tenant session-leakage vulnerability** across the Onboarding and Teaching applications, restoring correct tenant, session, and domain isolation.
- Built and documented an internal service SDK and provisioned infrastructure via AWS CDK across **5 microservices**.

## Projects

**Linkora -- Scalable URL Shortener** *Nuxt.js, Vue.js, TypeScript, Express.js, DynamoDB, AWS CDK*

**Live:** linkora-mnyr.onrender.com | **Code:** github.com/imsamyak/url-shortner | **Architecture:** linkora-mnyr.onrender.com/architecture

- Engineered a **full-stack URL shortener** with a **Nuxt 4/Vue 3 SSR client** and a modular **Express 5/TypeScript REST API**; implemented authenticated workspaces, link creation, expiration, and redirection.
- Designed a **single-table DynamoDB model** with **cursor-based pagination** and **native TTL attributes** for scalable access patterns and automatic cleanup of expired redirects.
- Secured API flows with **password hashing**, **intent-scoped authentication**, **schema validation**, and **protected server-side gateways**.
- Provisioned a **two-AZ cloud architecture** with traffic distribution, auto scaling, firewall protection, and monitoring, plus rolling CI /CD; tested core services with unit and end-to-end tests.

## Education

**Maharaja Agrasen Institute of Technology, GGSIP University — Delhi**

*Bachelor of Technology in Computer Science and Engineering — CGPA: 8.95* *2025*

## Technical Skills

**Languages:** Java, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

**Frontend & UI:** React.js, Next.js, Redux, Tailwind CSS, SCSS

**Backend & APIs:** Node.js, Express.js, Spring Boot, NestJS, JPA, Hibernate, Spring MVC, REST, OpenAPI, GraphQL

**Databases:** PostgreSQL, MySQL, DynamoDB

**Messaging:** Kafka, AWS SQS, AWS SNS

**Cloud & DevOps:** AWS Lambda, AWS S3, CloudWatch, Docker, CI/CD

**Tools & Testing:** Git, GitHub, Postman, SonarQube, Vitest, Playwright

**Concepts:** SOLID Principles, API Design, Agile Methodologies, Data Structures & Algorithms, OOPs, Microservices, Event-Driven Architecture, Domain-Driven Design, CQRS, Cloud Computing, Core Web Vitals

**AI-Assisted Development:** GitHub Copilot, Claude

## Certifications

- Java Full Stack Development -- Wipro TalentNext